THE DERIVED INTERNAL HOM ALONG A FINITE RING MAP PRESERVES BOUNDED BELOW COHERENT COMPLEXES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The identification of the derived internal hom along a ring map with the Ext groups of the base ring. Invoked inside the counted proof (source line 2489). Turns the finiteness of the cohomology of the internal hom into the finiteness of Ext modules of the base ring.
- The finiteness of Ext over a Noetherian ring between a finite module and a finite module. Invoked inside the counted proof (source line 2498). Supplies the finiteness of the spectral sequence's E2 terms.

Source reference. Original lemma lemma-RHom-ext: sources/stacks-dualizing/pinned-original-source.tex, source lines 2452–2452. Navigation pointer only: the counted text refers to the Stacks item lemma-RHom-ext, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference algebra-lemma-ext-noetherian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1958–1958. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-ext-noetherian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference derived-section-triangulated-sub: sources/stacks-dualizing/pinned-original-source.tex, source lines 1363–1363. Navigation pointer only: the counted text refers to the Stacks item derived-section-triangulated-sub, which belongs to another chapter of the project and is not part of this card. The

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Lemma 0.1. *With notation as above, assume $A \to B$ is a finite ring map of Noetherian rings. Then $R\mathrm{Hom}(B, -)$ maps $D^+_{Coh}(A)$ into $D^+_{Coh}(B)$.*

Let A be a Noetherian ring. We will denote

$$D_{Coh}(A) \subset D(A)$$

the full subcategory consisting of those objects K of $D(A)$ whose cohomology modules are all finite A-modules. This makes sense by Derived Categories, Section derived-section-triangulated-sub because as A is Noetherian, the subcategory of finite A-modules is a Serre subcategory of Mod_A.

Proof. We have to show: if $K \in D^+(A)$ has finite cohomology modules, then the complex $R\mathrm{Hom}(B, K)$ has finite cohomology modules too. This follows for example from Lemma lemma-RHom-ext if we can show the ext modules $\mathrm{Ext}^i_A(B, K)$ are finite A-modules. Since K is bounded below there is a convergent spectral sequence

$$\mathrm{Ext}^p_A(B, H^q(K)) \Rightarrow \mathrm{Ext}^{p+q}_A(B, K)$$

This finishes the proof as the modules $\mathrm{Ext}^p_A(B, H^q(K))$ are finite by Algebra, Lemma algebra-lemma-ext-noetherian. □